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CAPSTONE ASSIGNMENT - 1 (COA)

CASE STUDY - 1 (Smart traffic Control System)

Ans 1
- Input Unit => collects data from sensors (traffic density signals).
- ALU => Processes data and performs calculations
- Control Unit => Makes decision (Signal timing control).
- Memory Unit => Stores data and instruction
- Output Unit => Sends signal to traffic light.

Ans 2 RISC is better because:
- Faster execution
- Simple instruction
- Efficient for real-time systems

use RISC architecture.

Ans 3 IPC and Performance
- IPC (Instruction per cycle) => Number of instruction executed per cycle.
- Higher IPC => faster processing -> quick response to traffic changes.
- Improves overall system performance and efficiency

Ans 4 Processor Type:
- A Pipeline processor is best because it can execute multiple instructions at the same time in different stages. This increases speed and improves real-time response, which is

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important for traffic control:
Single cycle is slower, and multi cycled is better than single but still less efficient than pipelining.

Ans 5 Hazards in pipelining

- Data hazards: when one instruction depend on another → solved by forwarding or stalling.
- Control Hazards: Due to branching → solved by branch prediction.
- Structural Hazards: when resources are limited → solved by better hardware or scheduling

These techniques help maintain smooth and fast system performance.

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CASE STUDY - 2 (Designing a small educational RISC-V Based Processor).

Ans 1 Importance of CPU registers Instruction Format and Decoding:

- **CPU Registers:**
  Small, fastest memory inside CPU (RISC-V has 32 registers). They store operands and results, reducing memory access time and improving speed.

- **Instruction format:**
  Defines how an instruction is arranged (opcode, source/destination registers, immediate values). RISC-V uses simple formats (R, I, S etc). making hardware design easier.

- **Instruction Decoding**
  The control unit interprets the instructions and generates signals for ALU, memory and registers to perform the correct operation

Ans 2 Basic instruction cycle
(example: ADD x1, x2, x3

1 **Fetch:** Instruction is fetched from memory using PC.

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2 Decode: CPU identifies operation and registers

3 Execute: ALU performs x2 + x3

4 Memory Access: Not required for ADD

5 Write Back: Result stored in register x1.

Ans 3 Addressing modes vs instruction types

- Addressing modes:
  - Register => Data in register (ADD x1, x2, x3)
  - Immediate => Constant value (ADD x1, x2, x3)
  - Base/Memory => Access memory (LW x1, 0(x2))

- Instruction Types:

  - Arithmetic (ADD, SUB)
  - Logical (AND, OR)
  - Load / store (LW, SW)
  - Control / Branch (BEQ, JAL)

These make execution flexible and efficient.

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Ans 4 Why RISC-V is suitable for education.

- Simple and easy to learn and understand ISA.
- Open-Source and free (No licensing)
- Fixed instruction formats → easier decoding
- Modular design → can add features step by step.
- Helps students clearly understand CPU working compared to complex ISAs.

Ans 5 Example of simple datapath Block diagram

- Main components:

* Program counter (PC)
* Instruction Memory
* Control unit.
* Register file
* ALU
* Data Memory.

Flow:

PC → Instruction memory → Decode (control unit) → Registers → ALU → Data Memory (if needed) → write back to register.

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